Polynomials: Combine and Distribute

Combine Like Terms

Coefficient ax^{n} Exponent

Variable(s)

'Like' terms must have
- **same variable(s)**
- **same exponent(s)**

Combine by
- **adding/subtracting coefficients of like terms**

Example: $3x^2y^3 + 10xy - 5x^2y^3 - 20xy + 4 = \boxed{-2x^2y^3 - 10xy + 4}$

***Ask:* How Many?**

x^2y^3	xy	x^0 (constants)
$3x^2y^3 - 5x^2y^3 = \boxed{-2x^2y^3}$	$+10xy - 20xy = \boxed{-10xy}$	$\boxed{+4}$

Distributive Property

$$a(b+c) = ab + ac$$

- **Multiply each term in the <u>second</u> polynomial by <u>every</u> term in the <u>first</u> polynomial**
- **Combine like terms to simplify**

Monomials: $a(b+c+d+e) = ab + ac + ad + ae$

Example: $-3x(7x^2 + 4x - 12) = \boxed{-21x^3 - 12x^2 + 36x}$

Polynomials: $(a+b)(c+d) = ac + ad + bc + bd$

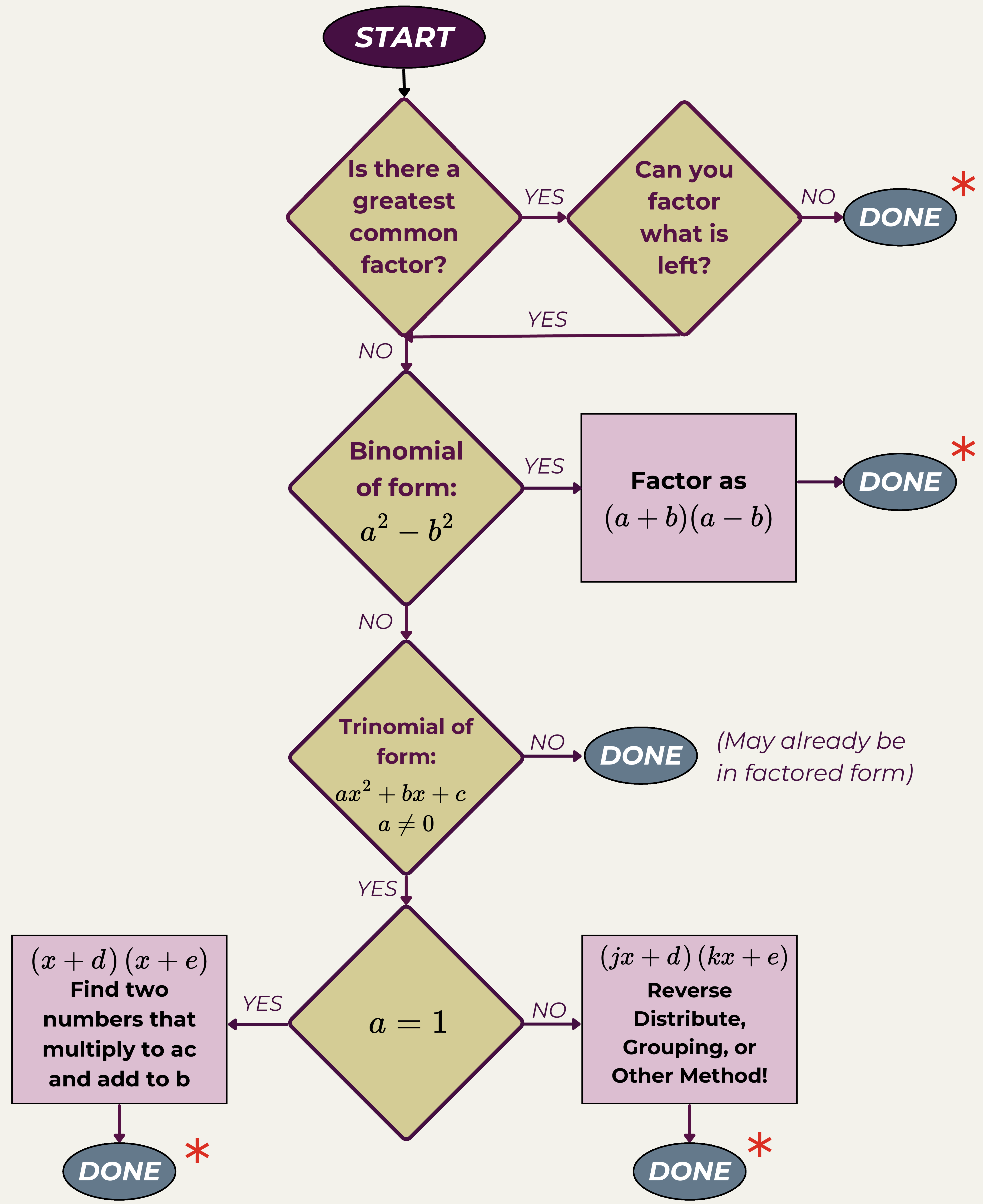
Examples:

$$(2x-3)(5x-8) = 10x^2 - 16x - 15x + 24$$
$$= \boxed{10x^2 - 31x + 24}$$

$$(x+y)(2x-3y+z) = 2x^2 - 3xy + xz + 2xy - 3y^2 + yz$$
$$= \boxed{2x^2 - xy + xz - 3y^2 + yz}$$

Quadratic Factoring Flowchart

Steps to factor binomial and trinomial quadratic expressions



*When you are **DONE**, double check to make sure that there are ***no common factors*** AND ***verify*** by multiplying!